

Lesson 7: Loops Practice

45 minutes

Overview

Students practice the basics of loops including using while loops, for loops, and updating multiple screen elements with a for loop. Along the way students develop debugging practices with loops.

Standards

Full Course Alignment

CSP Conceptual Framework

- **AAP-2** - The way statements are sequenced and combined in a program determines the computed result. Programs incorporate iteration and selection constructs to represent repetition and make decisions to handle varied input values.
- **CRD-2** - Developers create and innovate using an iterative design process that is user-focused, that incorporates implementation/feedback cycles, and that leaves ample room for experimentation and risk-taking.

CSTA K-12 Computer Science Standards (2017)

- **AP** - Algorithms & Programming
- **DA** - Data & Analysis

Agenda

Warm Up (5 minutes)

Quick Warm Up

Activity (35 minutes)

Practice Time

Wrap Up (5 minutes)

Assessment: Check For Understanding: AP Practice

Objectives

Students will be able to:

- Debug programs that use loops
- Use a for-loop to update multiple screen elements at once
- Write programs that use for loops with the support of sample code.

Preparation

- Review the programming progression to understand what students will be asked to do
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Unit 6 - Lesson 7 - Loops**

Practice - Slides [▼ Make a Copy](#)

Teaching Guide

Warm Up (5 minutes)

💡 Teaching Tip

Move Quickly to the Activity: There's a lot in the main activity of today's lesson. You may optionally wish to do a quick vocabulary review or address any questions that came up in the last lesson. Otherwise, give students more time to get hands on with some code.

Quick Warm Up

Remarks

Today we're going to have a chance to practice programming with a lot of the concepts and patterns we've explored over the last two lessons. I encourage you to go through these with a partner, but pay close attention to what each other is doing. In our next lesson you're going to have to use a lot of these on an independent project, and these activities are good practice for what you'll find there! Alright, let's get to it!

Activity (35 minutes)

Practice Time

Teaching Tip

Providing Support: Circulate around the room through the lesson encouraging students to use the strategies introduced at the beginning of the lesson. Students have a number of supports at their fingertips, so a big part of your role is helping build their independence in using those resources.

Remarks

 Today you're mostly going to practice what we've learned about programming with loops. I'm here to help you when you need. However, I first want to remind you of the following:

- Use your debugging skills. Try to zoom in on precisely where you're getting stuck.
- Talk to your partner! That's what they're there for!
- Hover over blocks to read the documentation about how they work.
- Read the resources in the Help & Tips tab
- Talk to the group next to you. If another group asks for help make sure to take some time to talk it through with them.

 We can debug loops by using the Watch Panel with our iterator variable, usually an `i`.

- Watcher Panel
 - Here you can see the length of a list in addition to all of the elements.

 **Do This:** Direct students to Lesson 3, Level 2 on Code Studio.

Levels 1-3 While Loop Practice: Simple `console.log` levels in which students practice using while loops.

- Level 1: Students use a while loop to print the numbers 0-99
- Level 2: Students use a while loop to print a message 100 times
- Level 3: Students use a while loop to fill an array and print the results
- Level 4: Students complete a choice level.



1-3

While Loop Practice

1

2

3



4

Putting It Together

Levels 5-7 For Loop Practice: Simple `console.log` levels to practice using for loops.

- Level 5: Students use a for loop to print the numbers 0-99
- Level 6: Students use a for loop to print a message 100 times
- Level 7: Students use a for loop to fill an array and print the results



5-7

For Loop Practice

5

6

7



Assessment Opportunity

**Create Performance Task Written Response Practice**

In order to help prepare students for the new component of the AP Create Performance Task, there will be several opportunities throughout the curriculum to practice creating written responses aligned with the learning objectives that might be assessed on the exam.

Sample Prompt: *Identify the number of times the body of your iteration statement will execute. Describe a condition or error that would cause your iteration statement to not terminate and cause an infinite loop. If no such condition or error exists, explain how the loop could be modified to cause an infinite loop.*

Unit Journal Template**Sample Response for Instructors**

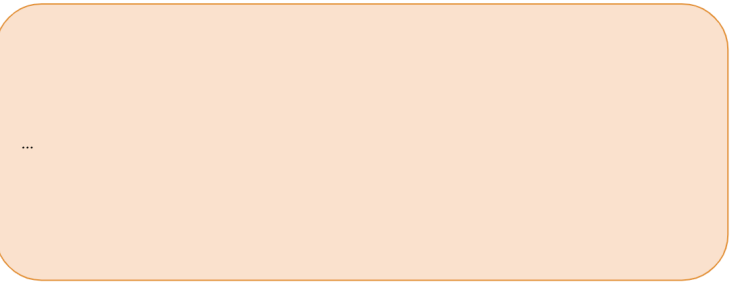
Programming Concept: *Loops*

Snip and paste an example code with loops here!

Code Highlights- Use these colored spotlights to point out & discuss features of this code



Identify the number of times the body of your iteration statement will execute. Describe a condition or error that would cause your iteration statement to not terminate and cause an infinite loop. If no such condition or error exists, explain how the loop could be modified to cause an infinite loop.



Code Highlights



Code Highlight 1

...



Code Highlight 2

...

Levels 8-9 Loops and Screen Elements: In these levels students practice writing programs that modify screen elements with loops. Screen elements have already been designed to have repeated patterns (e.g. "dice0", "dice1", "dice2" ...) that make it possible to write code that modifies every screen element. This is important practice for the Make project in the next lesson.

- Level 8: This level is a simple introduction to using loops to generate a sentence with random font colors and font sizes.
- Level 9: Students must update a "Dice Roller" app that has some working functionality already. All of the different concepts they'll need to use are already in the starter code of this app, but they'll need to think through how to put them together to get the target code behavior.



8-9

Loops and Screen Elements

8

9



Assessment Opportunity ▲

Create Performance Task Written Response Practice

In order to help prepare students for the new component of the AP Create Performance Task, there will be several opportunities throughout the curriculum to practice creating written responses aligned with the learning objectives that might be assessed on the exam.

Create PT Learning Objective(s):

- Identify the error.
- Correct the error.

Unit Journal Template You may choose to reuse this debugging journal template if you think it will help support your students becoming stronger debuggers.

Sample Response for Instructors



Write about a bug you found during your work in class



Describe the problem

- What did you expect it to do?
- What did it actually do?
- Did it always happen?

Write your response here...

Hunt for bugs

- What caused the bug?
- How did you find the bug?

Write your response here...

Try Solutions

- What changes did you make?
- What was the solution you found?

Write your response here...

Wrap Up (5 minutes)

Discuss: *What aspects of working with loops do you feel like clicked today? What do you still feel like you have trouble with?*

Discussion Goal: Use this opportunity to address any lingering questions or misconceptions in the room. You can also use this as a source of discussion topics to kick off the following lesson. As you lead the discussion, call out the many resources students have access to help when they're getting stuck.

Have students share with one another before sharing with the whole class.



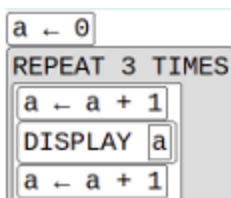
Remarks

Working with loops can be tricky - especially working with the iterator variable. We will get more practice tomorrow by making an app that uses loops in an interesting way.

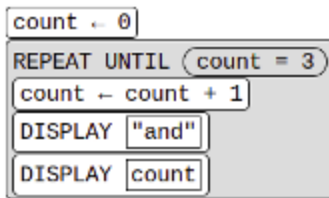
Assessment: Check For Understanding: AP Practice

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: What will be displayed after this code segment is run?



Question: What will be displayed after this code segment is run?



☰ 10-11

Check For Understanding: AP Practice



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